OWNER'S MANUAL · COILOVER KIT

Bilanx Owners Manual

Installation, ride height, damper adjustment and setup data.

Issued 2026-11-07 Adjuster 0–32 clicks Shaftworks USA · Chino, CA

01 KIT CONTENTS

DESCRIPTION	PCS
Front coilover assembly	2
Rear coilover assembly	2
Spanner wrench	2
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NOTE
Before you install this product, check the kit contents. If anything is missing, contact Shaftworks USA.

NOTE
There can be small differences between your product and the images in these instructions.

WARNING
Read this manual before you install this product. Suspension is an important part of the vehicle and its stability.

02 MOUNTING INSTRUCTIONS

WARNING
We strongly recommend having this product installed by a qualified shop.

WARNING
If you work with a lifted vehicle, make sure it is safely supported so it cannot tip over.

NOTE
See the vehicle service manual for vehicle-specific procedures and torque values.

1. Raise the vehicle and put it on jack stands at the factory lift points.
2. Remove the wheels.
3. Remove the upper and lower attachments and remove the original shock absorbers.
4. Install the Bilanx coilovers on the vehicle.
5. Fasten the upper and lower attachments. Tighten to the factory torque values.
6. Secure brake hoses and ABS wires so they cannot touch the coilover.
7. Reinstall all removed parts the same way they were before installation.
8. Set ride height and check bump travel and droop before driving (see Adjustments).
9. Have the vehicle aligned.

NOTE
Make sure all bolts are tightened to the correct torque, and that nothing fouls or restricts the coilover at full compression, full extension and full steering lock.

03 ADJUSTMENTS

> ⚠ **WARNING**
> Before you drive, make sure the setup matches the recommended setup data.
> Contact Shaftworks USA if you have any questions about setting up.

Your recommended damping and spring rates are on the setup sheet in the box.

Check your dampers for a "Do not adjust" sticker.

`DO NOT ADJUST` **STICKER PRESENT**

Damper length has been set for your car. Do not move the lower mount. Set ride height with the spring seat only.

NO STICKER

Damper length is not pre-set. You must check bump travel and droop for your car before driving.

Adjust in this order: **ride height and travel, then damping, then tuning.** Re-check alignment after any height change.

RIDE HEIGHT AND TRAVEL

Ride height is set with the **spring seat**. Damper length is set at the **lower mount** and controls how travel is split between bump (compression) and droop (extension). A loose or preloaded spring at your chosen height is normal and does not change how the car handles.

1. Loosen the lock ring with the supplied spanner.
2. Wind the spring seat up to raise the car, or down to lower it. Match left and right.
3. Tighten the lock ring against the seat.
4. Settle the car (roll it back and forth), then measure hub center to fender lip at each corner.

Check droop: keep at least `50 mm` below ride height

1. Measure tire bottom to fender at ride height.
2. Lift both wheels on that axle off the ground and measure again.
3. The difference is your droop.

Check bump travel: keep at least 35 mm at ride height

1. At ride height, slide the bump stop down the shaft until it touches the damper body.
2. Fit a zip tie on the shaft against the seal.
3. Drive the car normally (or a few laps), then measure how far the zip tie was pushed up. That is the travel you used.
4. If the bump stop is fully compressed in normal driving, you need more bump travel.

> **UNDER 35 MM WITH DAMPER LENGTH ALREADY SET?**
> With a "Do not adjust" sticker, or the lower mount at the groove limit, you may need a stiffer spring rate for that ride height. Contact us for a recommendation.

No sticker? Balance bump and droop at the lower mount

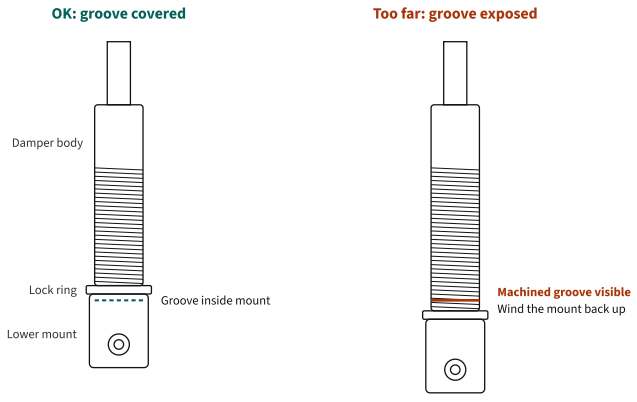
Then reset ride height with the spring seat.

CHANGE	BUMP TRAVEL	DROOP	THEN
Lower mount down (damper longer)	Less	More	Spring seat down the same amount
Lower mount up (damper shorter)	More	Less	Spring seat up the same amount

Example: you have 25 mm of droop and want 50 mm → lower mount down 25 mm, spring seat down 25 mm. Ride height stays the same. Re-check bump travel after.

Minimum thread engagement

It is marked by the machined groove on the damper body. Never wind the lower mount down far enough to expose the groove. Re-check bump travel and droop after every ride height change.



Lower mount limit: the machined groove on the damper body must stay covered.

DAMPER ADJUSTMENT

The adjuster on top of each damper has **about 32 clicks** of range. Clockwise is firmer, counterclockwise is softer, and it stops at both ends.

- **Count from full firm.** Turn clockwise gently until it stops: that is `0`. Count clicks counterclockwise from there (1, 2, 3… to about 32, full soft).
- **Start from the setup sheet.** Your recommended setting for each corner is on the setup sheet. Return to it whenever you lose track.
- **Change one end at a time,** 2 clicks at a time, and drive the same road or track to compare.
- **Keep left and right the same** unless you are corner-tuning on purpose.
- **Never force the adjuster** past its stop. It is a needle valve and can be damaged.

BILANX ONE, P2 AND P3 ADJUSTERS

Bilanx dampers come in three levels. Each step up adds an adjuster, so you can tune more of the damper's behavior separately.

Bilanx ONE

SINGLE ADJUSTABLE

1 adjuster

Damping — **32** clicks
Top of damper

Bilanx P2

DOUBLE ADJUSTABLE

2 adjusters

Rebound — **32** clicks
Top of piston rod

Compression — **22** clicks
Reservoir

Bilanx P3

TRIPLE ADJUSTABLE

3 adjusters

Rebound — **32** clicks
Top of piston rod

Low-speed compression — **25** clicks
Reservoir

High-speed compression — **17** clicks
Reservoir

Rebound
How fast the damper extends: body control and how quickly the wheel returns.

Compression
Resistance to compressing.

Low-speed compression
Slow movements: roll, dive, squat and weight transfer.

High-speed compression
Fast movements: bumps, curbs and sharp edges.

1. Start from the settings on your setup sheet.
2. Count every adjuster from full firm, the same as the single adjustable. Turn clockwise gently until it stops: that is **0**. Count clicks counterclockwise from there.
3. Never force a knob past its stop at either end.
4. Set rebound first, then compression.
5. Change one adjuster, at one end of the car, 1–2 clicks at a time, and drive the same road or track to compare.
6. Keep left and right the same unless you are corner-tuning on purpose.
7. Write down every change.

P3 TUNING TIPS

If the car rolls, dives or squats too much, add low-speed compression. If it feels harsh or skips over bumps and curbs, soften high-speed compression.

NOTE

Route reservoir hoses so they cannot touch the tire, brakes or exhaust through full travel and full steering lock. Secure them with zip ties.

REMOTE RESERVOIR QUICK DISCONNECT

WARNING

Never drive the car or cycle a damper with its reservoir disconnected. This damages the damper internals.

WARNING

The reservoir is charged with high-pressure nitrogen. Release all pressure before disconnecting.

DISCONNECTING

1. Unload the damper. Put the car on jack stands and disconnect the anti-roll bar so the wheels hang at full droop, or remove the damper from the car.
2. Release all nitrogen pressure through the Schrader valve on the reservoir. Make sure no pressure is left.
3. Disconnect the reservoir at the quick disconnect. Keep both sides of the fitting clean.

RECONNECTING

1. Check both sides of the fitting are clean, then connect the reservoir fully.
2. With the damper at full droop, charge the reservoir with nitrogen only (never air) to **180 psi** using a gauge.
3. The damper is ready to use.

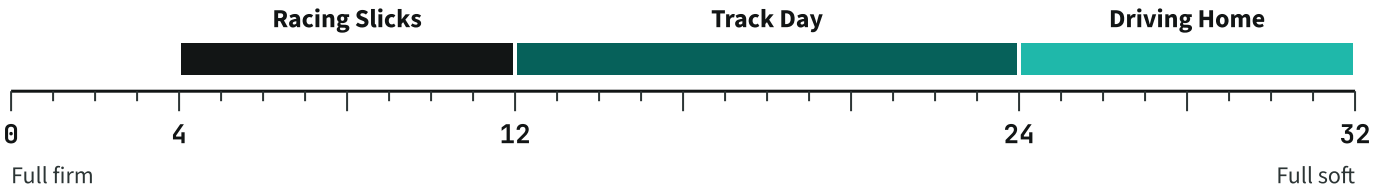
04 SETUP DATA

0–32	35 mm	50 mm
Adjuster range, clicks from full firm	Minimum bump travel at ride height	Minimum droop at ride height

Lower mount limit	Machined groove on damper body must stay covered
Spring rates and recommended damping	See setup sheet

REBOUND STARTING POINTS

Clicks counterclockwise from full firm (0) to full soft (32).



USE	REBOUND (CLICKS)
● Racing Slicks	4–12
● Track Day / Performance Driving	12–24
● Driving Home	24–32

> **NEED 0-4 CLICKS?**
> If you need to run 0–4 clicks, contact Shaftworks USA about revalving.

RECOMMENDED SERVICE

USE	SERVICE INTERVAL
Competition vehicles (optimal, consistent performance)	Every 2 years
Racing	As often as desired; 2–5 years is expected
Street use	Every 18,000 miles or 5 years, whichever comes first

Contact Shaftworks USA to schedule a rebuild.

05 TUNING GUIDE

Fix handling one change at a time, starting with the first item in each row.
Understeer means the front slides wide; oversteer means the rear steps out.

IF THE CAR…	TRY (IN ORDER)
Understeers on corner entry	Softer front damping · firmer rear damping · raise rear height · firmer rear bar
Understeers mid-corner / exit	Firmer rear damping · firmer rear bar · stiffer rear spring
Oversteers	Softer rear damping · lower rear height · softer rear bar · stiffer front spring
Lacks traction under power	Softer damping on the drive wheels · softer bar on the drive axle
Unsettled under braking	Softer front damping · firmer rear damping
Hits bump stops or bottoms out	More ride height · firmer damping · stiffer spring (last resort)
Feels harsh over bumps	Soften damping 2 clicks at a time on the harsh end

STILL NOT RIGHT?
Contact Shaftworks USA at **sales@shaftworksusa.com** with your settings and a description of the problem.

Shaftworks USA products are subject to continuous improvement and development. These instructions include the most up-to-date information available at the time of issue, but minor updates may occur.

For the latest information or any questions about this document, contact Shaftworks USA.

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